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Interpretable Statistical Modelling for Real-Time Sepsis-3 Onset Prediction in MIMIC-IV, with External Validation on eICU-CRD

By
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Abstract

Sepsis is a leading cause of intensive-care mortality, and earlier recognition improves survival. Research on sepsis prediction is dominated by machine learning and, on the MIMIC-IV database specifically, by prognostic modelling of mortality *after* diagnosis rather than prediction of *onset* itself. This thesis develops and validates a fully interpretable, purely statistical pipeline for real-time Sepsis-3 onset prediction. Using 65,241 adult ICU stays from MIMIC-IV v3.1, each stay was resampled to an hourly grid of vitals and laboratory values, hourly Sequential Organ Failure Assessment (SOFA) scores were computed, and the exact onset hour was labelled under the Sepsis-3 consensus definition. Continuously updated risk was obtained through a dynamic landmark supermodel rather than any neural or ensemble method: a single elastic-net logistic model fitted on stacked landmark datasets across six prediction hours, achieving a pooled out-of-sample AUROC of 0.797.

Three methodological commitments distinguish the work. First, the proportional-hazards assumption underlying Cox regression was formally tested using Schoenfeld residuals and, because the global test failed, a time-varying reformulation was fitted in response. Second, incident (new-onset) sepsis was separated from prevalent sepsis using an hour-six landmark, which revealed that standard bedside scores range from non-discriminative to inverse on this task (SOFA area under the receiver operating characteristic curve 0.36, qSOFA 0.49, SIRS 0.59), whereas a multivariable landmark nomogram attained 0.775. Third, an informative-missingness analysis showed that whether a laboratory test was ordered is a stronger predictor than its measured value, quantifying a label-recognition ceiling intrinsic to retrospectively labelled ICU data. Beyond this internal analysis, the frozen model was externally validated on the eICU Collaborative Research Database of more than 200 United States ICUs: discrimination transported with modest degradation (area under the curve 0.686 against an internal eICU ceiling of 0.711), and a single-parameter recalibration corrected the expected miscalibration. All extraction, modelling, and evaluation code is released for reproducibility.

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This research used the MIMIC-IV and eICU Collaborative Research Database resources made available through PhysioNet by the MIT Laboratory for Computational Physiology and their partner institutions. I am grateful to the data custodians and to the credentialing process that makes responsible secondary use of critical-care data possible.

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Chapter 1

Introduction

1.1 Context and Motivation

Sepsis, defined by the Sepsis-3 consensus as life-threatening organ dysfunction caused by a dysregulated host response to infection [1], remains a leading cause of mortality in the intensive care unit (ICU). The Global Burden of Disease study attributed an estimated 11 million deaths to sepsis in 2017, nearly 20% of all deaths worldwide [2]. Because each additional hour before effective treatment is associated with higher mortality [3], predicting the *onset* of sepsis ahead of clinical recognition is of direct clinical value: an early warning, even by a few hours, can accelerate the initiation of antibiotics, fluid resuscitation, and source control.

Two modelling traditions address this prediction problem. The first, and by far the more actively published, is machine learning: gradient-boosted trees, recurrent neural networks, and transformer architectures dominate the recent literature. The second, older and still foundational, is classical statistics: Cox proportional-hazards regression, logistic regression, generalised estimating equations, and trajectory modelling. Statistical models retain specific advantages for real-time clinical use that are often undervalued in the current literature:

- They yield directly interpretable hazard ratios and odds ratios that clinicians can relate to their bedside experience.
- Their assumptions (proportional hazards, linearity in the log-odds, independence of observations) can be formally tested, and the consequences of violation can be quantified and addressed.
- They can be benchmarked transparently against the established rule-based scores that clinicians already use: the Systemic Inflammatory Response Syndrome (SIRS) criteria, the quick Sequential Organ Failure Assessment (qSOFA), and the full SOFA score.

- Their coefficients are closed-form and portable: a model fitted on one database can be transported to another by applying the same linear predictor, without retraining or access to the development data.

On the MIMIC-IV database [4] (a large, de-identified critical-care database from the Beth Israel Deaconess Medical Center containing over 65,000 adult ICU stays), the statistical literature has developed unevenly. As detailed in Chapter 3, most studies predict mortality *after* sepsis is diagnosed rather than onset itself; the proportional-hazards assumption is rarely tested before a Cox model is fitted; and external validation is largely absent from the onset-prediction literature specifically. The single statistical real-time onset benchmark, TREWScore [5], was fitted on the older MIMIC-II database in 2015 and has not been revisited on MIMIC-IV with purely statistical, interpretable tools.

1.2 Research Gap

Three specific gaps motivate this work and are established in detail through the systematic review of Chapter 3:

- **G1 (Onset versus prognosis).** MIMIC-IV statistical work overwhelmingly predicts mortality after diagnosis. The dominant pattern couples LASSO variable selection with a logistic nomogram [6], and the trajectory literature uses GBTM to link SOFA shapes to mortality [7]. Genuine pre-onset, real-time detection with purely statistical tools has not been revisited on MIMIC-IV since TREWScore’s 2015 MIMIC-II study [5].
- **G2 (Untested proportional hazards).** Sepsis hazards are frequently non-proportional [8], yet no reviewed MIMIC-IV Cox study reports testing the assumption, weakening confidence in the reported hazard ratios. The appropriate response to a failed proportional-hazards test, a time-varying reformulation, is absent from the MIMIC-IV onset literature.
- **G3 (Validation asymmetry).** External validation is common in the prognostic nomogram literature but essentially absent from onset-prediction and trajectory studies. Wang et al. [9] reviewed 91 sepsis real-time prediction models and found that the median AUROC falls from 0.886 internally to 0.783 externally, demonstrating that the internal-to-external gap, rather than raw internal discrimination, is the field’s central weakness. This gap is precisely where onset-prediction models are most vulnerable, yet precisely where validation is most lacking.

1.3 Research Question and Objectives

These gaps motivate one primary research question and three sub-questions.

RQ. *Can a purely statistical (Cox proportional-hazards / time-varying / landmark) model, using routinely available real-time vitals and laboratory values in MIMIC-IV, predict impending Sepsis-3 onset ahead of clinical recognition, with performance comparable to or exceeding the rule-based scores SIRS, qSOFA, and SOFA?*

- **SQ1.** Which routinely measured variables are the strongest statistically significant predictors of incident onset, and does the proportional-hazards assumption hold for the fitted Cox model?
- **SQ2.** Does modelling trajectories through a time-varying or dynamic landmark extension improve early detection over a static single-time-point model?
- **SQ3.** How does discriminative performance compare with SIRS, qSOFA, and SOFA on the correct incident-onset task, and does it survive the transition from internal (MIMIC-IV) to external (eICU-CRD) evaluation?

1.4 Contributions

This thesis makes the following contributions, each mapped to the gap it closes:

1. **C1: First interpretable statistical real-time Sepsis-3 onset model on MIMIC-IV (closes G1).** A dynamic landmark supermodel [10], [11] is fitted across six landmark hours {6, 12, 18, 24, 36, 48} with feature-by-time interactions, yielding one interpretable model that emits continuously updated hourly risk. Incident sepsis is explicitly separated from prevalent sepsis using an hour-six landmark, and the model reaches an overall pooled out-of-sample AUROC of 0.797 across all landmarks.
2. **C2: Explicit proportional-hazards testing with a triggered time-varying reformulation (closes G2).** The Schoenfeld residual test is applied to the onset Cox model; because the global test fails ($p < 10^{-16}$), a time-varying Cox model on a counting-process formulation is fitted in direct response, demonstrating a workflow that the MIMIC-IV literature has omitted.
3. **C3: Reframing the benchmark.** On the correct incident-onset target, standard bedside scores range from non-discriminative to inverse: SOFA at hour 6 achieves an AUROC of only 0.36, qSOFA 0.49, and SIRS 0.59. A

multivariable landmark nomogram attains 0.775, demonstrating that severity scores detect *established* organ failure, not *impending* onset.

4. **C4: Informative-missingness characterisation.** Binary “was this lab ordered” indicators are the strongest predictors in the elastic-net nomogram [12], [13]. The P/F ratio measurement indicator alone carries an odds ratio of 5.46, quantifying a label-recognition ceiling intrinsic to retrospectively labelled ICU data.
5. **C5: A reproducible, transportable pipeline that is externally validated on eICU-CRD (closes G3).** The frozen MIMIC model is applied without refitting to the eICU Collaborative Research Database [14] (113,597 at-risk stays, 200+ US ICUs). Discrimination transports at AUROC 0.686 against an internal eICU ceiling of 0.711, and a single-parameter recalibration corrects the expected miscalibration.

1.5 Thesis Structure

The remainder of this thesis is organised as follows. Chapter 2 sets out the research methodology, data sources, cohort construction, and evaluation framework. Chapter 3 reviews the related statistical and machine-learning literature and formalises the three gaps above. Chapter 4 describes the system architecture and the design of each pipeline stage, from data extraction through the seven statistical analyses. Chapter 5 presents and critically discusses the internal and external results, with particular emphasis on the dynamic landmark supermodel (Analysis 2), which is the central methodological contribution of this work. Chapter 6 concludes with a summary of findings and avenues for further research.

Chapter 2

Research Methodology

2.1 Research Approach

The study is a retrospective, observational modelling study on de-identified critical-care data, conducted as quantitative empirical research. The research question (Section 1.3) is operationalised as a prediction task with pre-registered baselines (SIRS, qSOFA, SOFA) and an explicit null hypothesis for the proportional-hazards assumption. A deliberate design choice constrains all modelling to classical statistics: this preserves interpretability, supports formal assumption testing, and establishes a transparent, transportable baseline against which machine-learning approaches can later be compared on identical data splits.

The methodology follows the standard prognostic-model development cycle recommended by the TRIPOD (Transparent Reporting of a multivariable prediction model for Individual Prognosis or Diagnosis) guidelines [15]: cohort definition, outcome labelling, feature construction, model fitting, internal validation, and external validation. Two overarching principles govern every stage.

Leakage safety. At any prediction time, only information that would genuinely be available to a clinician at that hour is used. This is implemented through forward-filled values (each variable retains its most recent measurement), time-since-last-measurement counters (encoding the staleness of information as an explicit signal), and strict temporal conditioning (the landmark design ensures that features are drawn only from the observation window preceding the prediction point).

Honest evaluation. Discrimination is measured out of sample using patient-grouped cross-validation, ensuring that no patient contributes to both the training and the test set. The frozen model is additionally transported to a second, inde-

pendent database with no refitting, providing the most stringent form of external validation.

2.2 Software Development Methodology

The pipeline was developed incrementally as a sequence of numbered, single-purpose stages (Chapter 4), each writing an inspectable intermediate artefact that the next stage consumes. This staged design supports reproducibility and debuggability: any stage can be re-run in isolation, and a single configuration module is the sole source of truth for file paths, clinical item identifiers, antibiotic name lists, and Sepsis-3 thresholds. Data extraction and panel construction are implemented in Python (pandas); statistical modelling and reporting are implemented in R, with each analysis stage emitting a self-contained, reproducible report. All stochastic steps use a fixed random seed (486649), and the multi-gigabyte source tables are processed by streaming in bounded-memory chunks rather than being loaded whole.

2.3 Data Sources

2.3.1 Development Data: MIMIC-IV v3.1

The primary development data is MIMIC-IV v3.1 [4], a large, de-identified critical-care database comprising records from the Beth Israel Deaconess Medical Center (BIDMC) in Boston, Massachusetts. The database contains comprehensive clinical data including demographics, vital signs, laboratory measurements, medications, procedures, and clinical notes for over 65,000 adult ICU stays. It was accessed under the required PhysioNet credentialing agreement.

2.3.2 External Validation Data: eICU-CRD v2.0

External validation uses the eICU Collaborative Research Database v2.0 [14], a multi-centre collection spanning more than 200 United States ICUs. The eICU-CRD is independently collected, uses different electronic health record systems, and represents a broader range of clinical practices and patient populations than the single-centre MIMIC-IV. Both the geographic and institutional diversity of eICU-CRD make it a meaningful external test: if the model’s signal is real, it should survive the transition to this heterogeneous population.

2.4 Cohort Construction and Inclusion Criteria

An adult ICU cohort was constructed by applying the following inclusion criteria:

- Age ≥ 18 years at ICU admission.
- First ICU stay only (to avoid within-patient correlation from repeated admissions).
- Length of stay ≥ 4 hours (to ensure sufficient data for hourly feature construction).

This yielded **65,241 stays** in MIMIC-IV. The identical criteria were applied to eICU-CRD, yielding a comparable external cohort.

2.5 Outcome Definition: Sepsis-3 Onset

Sepsis-3 is defined following Seymour et al. [16] as suspected infection accompanied by an acute rise of two or more points in the SOFA score. Suspected infection is operationalised as a qualifying antibiotic order paired with a body-fluid culture within the Sepsis-3 time windows (culture within 24 hours before to 72 hours after the antibiotic, or antibiotic within 72 hours of culture). Surveillance swabs (e.g., MRSA screens) are excluded to avoid inflating the suspicion rate.

The cohort prevalence of Sepsis-3 on the first-24h analysis matrix is **37.1%**, with septic shock at 9.2% and in-hospital mortality of 18.7% in the septic group versus 6.1% in the non-septic group. These rates are consistent with a valid label and with the published literature. A slightly smaller fraction (32.0%) can be assigned an *exact hourly onset time* within the 72-hour panel; this temporal subset is the one carried forward into the landmark analyses, since it is the onset *hour*, not merely the label, that the real-time task requires.

2.6 Hourly Panel and Temporal Alignment

Every stay was resampled onto a regular hourly grid, yielding approximately **3.1 million stay-hours** across the 65,241 stays (mean 47 hours per stay). For each variable, three quantities are retained per hour:

1. The *raw value*: the measurement recorded at that hour, if any.
2. The *forward-filled value*: the most recent measurement, which is what the clinician currently knows.

3. The *time-since-last-measured* counter: the number of hours since the variable was last recorded, encoding irregular sampling as an explicit signal.

Hourly SOFA is scored from the forward-filled values using the six organ sub-systems (respiration, coagulation, liver, cardiovascular, central nervous system, and renal). The exact onset hour is located as the first hour inside the suspected-infection window at which the hourly SOFA rise reaches the two-point threshold. A per-hour binary label “onset within 12 hours” is then derived for early-warning evaluation.

2.7 The Landmark Design

The **hour-six landmark** is central to the study design. At hour 6 of each stay, the analysis conditions on the patient being still at risk (not yet septic), yielding **48,829 at-risk stays**. Of these, **4,750** (9.7%) develop incident sepsis within the subsequent observation window, and **3,330** (6.8%) within 12 hours. This is a realistically imbalanced early-prediction problem, unlike the near-balanced classification that a static cross-sectional analysis produces. It also separates the clinically actionable target (incident sepsis) from prevalent sepsis that is already present at admission and not predictable by any early-warning system.

For the dynamic landmark supermodel (Analysis 2), the landmark construction is repeated at hours {6, 12, 18, 24, 36, 48}, producing a stacked dataset of **207,010 stay-landmark rows** across 48,829 unique stays. The supermodel is fitted once on this stacked dataset with the landmark time s , its quadratic term s^2 , and all feature-by- s interactions as additional covariates, so that evaluating the model at the current hour yields continuously updated risk.

2.8 Statistical Models

The pipeline implements seven analytical stages, each addressing a specific methodological question. The first analysis establishes the discrimination ceiling; the second builds the real-time deliverable; analyses three through seven provide complementary inferential and evaluative evidence.

Analysis 1: Full-feature elastic-net nomogram. An elastic-net penalised logistic regression at the hour-six landmark uses the full feature set: vitals (heart rate, respiratory rate, MAP, SpO₂, temperature, GCS), severity scores (SOFA, qSOFA, SIRS), laboratory values (lactate, creatinine, WBC, P/F ratio), and six-hour trajectory slopes. Missing labs (70–88% missing at hour 6) are handled by an informative-missingness design: median imputation plus a binary “was measured”

indicator per lab and per slope. Evaluation uses nested cross-validation (5-fold outer, 5-fold inner for λ selection).

Analysis 2: Dynamic landmark supermodel (central contribution). The stacked landmark dataset is fitted as one penalised logistic model with s , s^2 , and 27 feature-by- s interactions, yielding 56 candidate features. Patient-grouped folds prevent information leakage across landmarks. The single model emits hourly-updating risk at any prediction hour, a genuine real-time deliverable built entirely from classical statistics.

Analysis 3: Within-patient correlation. Generalised estimating equations (GEE) with an exchangeable working correlation structure and cluster-robust (sandwich) standard errors quantify the within-patient correlation ignored by pooled models. A ridge-penalised time-varying Cox model on the counting-process formulation provides inferentially valid hazard ratios.

Analysis 4: Real-time evaluation. The supermodel is evaluated on a held-out set of 6,000 stays using the PhysioNet/CinC 2019 utility score and an alarm-burden analysis that trades sensitivity against false alarms per patient-day.

Analysis 5: Subgroup fairness. Performance is stratified by sex and age band along three axes: discrimination (AUROC), calibration-in-the-large, and error-rate parity (sensitivity and false-positive rate at a shared decision threshold), the last giving an equal-opportunity view of equitable performance across demographic subgroups.

Analysis 6: Group-based trajectory modelling. A latent-class growth model is fitted to first-24h SOFA trajectories, and class membership is linked to both mortality and incident sepsis via Cox regression and logistic regression respectively.

Analysis 7: Confounder-adjusted causal analysis. The effect of early antibiotic administration (≤ 3 hours) on in-hospital mortality among Sepsis-3 patients is estimated using four estimators of increasing robustness: crude unadjusted, propensity-score matching (PSM), inverse-probability weighting (IPW), and augmented IPW (doubly robust).

2.9 Evaluation Framework

Discrimination. The area under the receiver operating characteristic curve (AUROC) with DeLong 95% confidence intervals. Precision-recall AUC (PR-AUC) supplements AUROC for the imbalanced incident-onset task.

Calibration. The Brier score, calibration-in-the-large (mean predicted risk versus observed event rate), and the calibration slope (from a logistic recalibration model; slope = 1 indicates perfect calibration of the linear predictor's spread).

Clinical utility. The PhysioNet/CinC 2019 normalised utility score (0 = never-alarm baseline, 1 = per-hour oracle) and an alarm-burden curve plotting timely detection sensitivity against false alarms per patient-day across a range of risk thresholds.

External validation. The frozen MIMIC coefficients are applied without any refitting to the eICU-CRD hour-six landmark table. The frozen-transport AUROC is compared against an eICU-native elastic-net refit (the internal eICU ceiling) to separate loss of discrimination from miscalibration.

Chapter 3

Literature Review

This chapter reviews the statistical and machine-learning literature on sepsis prediction in the intensive care unit, traces the methodological lineage that informs the present work, and formalises three research gaps that the thesis addresses. The review is structured thematically: Section 3.1 introduces the clinical definition; Section 3.2 covers statistical onset-prediction models; Section 3.3 surveys prognostic nomograms and trajectory studies on MIMIC-IV; Section 3.4 reviews the machine-learning sepsis prediction landscape; Section 3.5 discusses landmarking and informative missingness; Section 3.6 reviews benchmarks and systematic reviews; and Section 3.7 synthesises the gaps.

3.1 The Sepsis-3 Definition and Its Operationalisation

The Sepsis-3 consensus [1] defines sepsis as life-threatening organ dysfunction caused by a dysregulated host response to infection, operationalised as suspected infection accompanied by an acute rise of two or more points in the Sequential Organ Failure Assessment (SOFA) score [17]. Seymour et al. [16] validated this operationalisation in large observational cohorts. Two features of this definition are methodologically consequential for prediction research. First, the label is *retrospective*: it depends on the clinician ordering cultures and antibiotics, so “onset” partly reflects the moment of clinical recognition rather than the true biological transition. Second, sepsis that is already present at ICU admission (*prevalent* sepsis) is not predictable by an early-warning system; only *incident* sepsis developing after admission is the appropriate target, yet many studies do not distinguish between the two.

3.2 Statistical Onset Prediction

The foundational statistical benchmark for real-time sepsis prediction is TREWScore [5], a Cox proportional-hazards model fitted on routinely collected MIMIC-II vitals and laboratory values. TREWScore achieved an area under the receiver operating characteristic curve (AUROC) of 0.83 with a median lead time of 28 hours before clinical recognition of septic shock. It demonstrated that a small set of physiological variables, modelled with a well-understood statistical framework, can anticipate clinical deterioration in real time. However, TREWScore was developed on MIMIC-II, which has since been superseded by two major database revisions. It has not been revisited on MIMIC-IV with purely statistical, interpretable tools.

Liaw et al. [18] proposed a longitudinal, multivariable survival model (a complementary log-log formulation) that tracks evolving mortality risk from repeatedly measured indicators, demonstrating the value of updating covariate values over the course of a stay rather than relying on a baseline snapshot. This time-varying approach is directly relevant to the current work, which extends the idea from mortality modelling to onset prediction. Kasal et al. [8] compared a standard Cox model against a flexible survival model in severe sepsis and found that hazards are frequently *non-proportional*, a methodological caution that has been largely ignored by subsequent MIMIC-IV studies. In the present review, no MIMIC-IV Cox study was found that reports testing the proportional-hazards assumption before interpreting hazard ratios, creating a gap in the validity of their reported effect sizes (**G2**).

3.3 Prognostic Modelling on MIMIC-IV

A well-established MIMIC-IV pattern couples LASSO variable selection with multivariable logistic regression to predict in-hospital mortality *after* a sepsis diagnosis has been made, typically presented as a nomogram [6]. These studies are often methodologically mature: they use penalised regression to guard against overfitting, report calibration curves and decision-curve analysis, and in several cases include external validation on the eICU Collaborative Research Database [14]. However, they address a fundamentally different clinical question, prognosis after diagnosis, rather than the early detection of onset itself.

Group-based trajectory modelling (GBTM) of SOFA scores represents a second strand of MIMIC-IV research. Yang et al. [7] fitted latent-class growth models to 72-hour SOFA trajectories and linked trajectory membership to mortality via Cox regression. Their approach captures the *shape* of organ-dysfunction evolution, not merely its endpoint, and demonstrated a strong dose-response relationship

between trajectory class and mortality. This trajectory-modelling thread informs the present work: however, Yang et al. [7] studied only mortality, and their analysis did not link trajectory classes to incident sepsis onset, nor did they test whether the trajectory shape carries predictive signal beyond its terminal SOFA value.

3.4 Machine-Learning Approaches to Sepsis Prediction

The machine-learning literature on sepsis prediction is substantially larger than the statistical literature and has been the subject of multiple systematic reviews. Gradient-boosted trees, recurrent neural networks, and transformer architectures have been applied to the PhysioNet/CinC 2019 Challenge data [19] and to MIMIC-IV directly, often achieving high internal discrimination. However, Fleuren et al. [20] conducted a systematic review and meta-analysis of machine-learning sepsis prediction models and reported large between-study heterogeneity in design, inclusion criteria, and reported performance. Several observations from this review are relevant:

- Internal AUROCs are often high but difficult to compare across studies because of differing outcome definitions, cohort construction, and temporal alignment of predictions to outcomes.
- Feature engineering, particularly the use of temporal features and laboratory ordering patterns, varies widely across studies, and the contribution of these engineered features versus raw measurements is rarely isolated.
- External validation is infrequent, and when performed, performance drops substantially.

A central argument of the present thesis is that interpretable statistical models retain specific advantages over black-box alternatives for clinical deployment: they yield directly interpretable hazard ratios and odds ratios, their assumptions can be formally tested and reported, and they can be benchmarked transparently against established rule-based scores. When the prediction ceiling is constrained by the labelling process rather than by model complexity, interpretability becomes a net gain rather than a cost.

3.5 Dynamic Landmarking and Informative Missingness

Dynamic prediction by landmarking [10] provides a principled framework for continuously updated risk prediction from survival data. The idea is simple but powerful: at each prediction time s (the “landmark”), the analysis is restricted to patients still at risk, their current covariate values are used, and a short-horizon outcome is modelled. The landmark supermodel formulation [11] extends this by stacking multiple landmark datasets and fitting a single penalised model with the landmark time s , its polynomial terms, and feature-by- s interactions as covariates. This yields one interpretable model that emits hourly-updating risk, rather than requiring a separate model at each prediction hour. To the author’s knowledge, this approach has not previously been applied to statistical Sepsis-3 onset prediction on MIMIC-IV.

A separate but complementary thread concerns informative missingness in electronic health records. Groenwold [12] formalised “the curse of knowing”: in clinical data, the fact that a test was ordered is itself informative, because clinicians order tests in response to suspicion. Singh et al. [13] showed empirically that binary “was this lab measured” indicators can outperform the measured values themselves as predictors of clinical outcomes. This phenomenon is particularly relevant to sepsis prediction, where laboratory tests such as lactate, creatinine, and the P/F ratio are ordered preferentially in sicker patients. An early-warning model that ignores missingness patterns discards a clinically meaningful signal.

3.6 Benchmarks, Validation, and Systematic Reviews

The PhysioNet/CinC 2019 Challenge [19] established a standardised early-prediction benchmark with a clinical-utility metric that rewards timely alarms and penalises both late detection and false alarms. This utility score provides a more clinically meaningful evaluation than AUROC alone, because it captures the temporal aspect of prediction: an alarm that fires six hours before onset is more valuable than one that fires at the moment of onset, and an alarm that fires on a non-septic patient imposes a real cost in clinician attention.

Wang et al. [9] conducted the most comprehensive methodological review of sepsis real-time prediction models to date, covering 91 studies. Their central finding is that the median AUROC falls from 0.886 under internal validation to 0.783 under full external validation. This degradation is consistent across model types and underscores that the internal-to-external gap, rather than raw internal dis-

crimination, is the field’s central weakness. Critically, this validation gap is most pronounced for onset-prediction models, precisely the task this thesis addresses. External validation is common in the MIMIC-IV prognostic nomogram literature but essentially absent from the onset-prediction and trajectory literature (**G3**).

3.7 Summary of Research Gaps

Synthesising the above, the three research gaps that structure this thesis are:

1. **G1 (Onset versus prognosis).** MIMIC-IV statistical work overwhelmingly predicts mortality after diagnosis. Genuine pre-onset, real-time detection with purely statistical tools has not been revisited on MIMIC-IV since TREWScore’s 2015 MIMIC-II study [5]. The dominant MIMIC-IV pattern (LASSO + nomogram) addresses prognosis, not onset [6]; and the trajectory literature (GBTM) links SOFA shapes to mortality, not to incident onset [7].
2. **G2 (Untested proportional hazards).** Non-proportional hazards in sepsis have been documented [8], yet no reviewed MIMIC-IV Cox study reports testing the assumption. When the assumption fails and is not addressed, the reported hazard ratios are unreliable; the appropriate response is a time-varying reformulation, which this thesis implements.
3. **G3 (Validation asymmetry).** External validation is common in the prognostic nomogram literature [6] but largely absent from onset-prediction and trajectory studies. Wang et al. [9] showed that this is precisely where performance degrades most, making external validation a necessary step rather than a desirable extra.

The contributions listed in Section 1.4 of Chapter 1 address each gap directly. Collectively, the gaps define a space for a purely statistical, interpretable, externally validated real-time sepsis onset model on MIMIC-IV, which is the contribution of this thesis.

Chapter 4

System Design

This chapter describes how the pipeline is built: the architecture that turns raw critical-care tables into a continuously updated, externally validated sepsis-onset risk. The design links directly to the best practices established in Chapter 3: leakage-safe temporal features, landmarking for real-time risk, informative-missingness encoding, and a transportable closed-form model.

4.1 Overall Architecture

The system is organised as a linear pipeline of numbered, single-purpose stages grouped into three layers (**Extract**, **Model**, and **Evaluate**), as shown in Figure 4.1. Each stage reads the artefact written by the previous stage and writes its own, so any stage can be re-run in isolation. A single configuration module (`config.py`) is the sole source of truth for file paths, clinical item identifiers, antibiotic name lists, and Sepsis-3 thresholds.

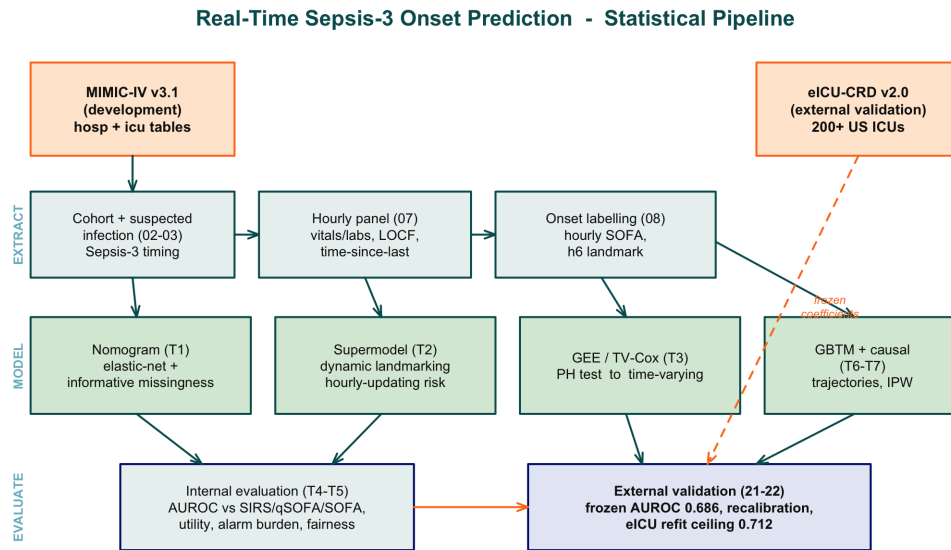


Figure 4.1: System architecture: the Extract, Model, and Evaluate layers. The development data (MIMIC-IV) flows through cohort construction, hourly panel assembly, and onset labelling into the seven statistical analyses; the frozen model coefficients are then transported to eICU-CRD for external validation.

4.2 Extract Layer (Scripts 01–08)

The extract layer builds the temporal substrate that the models consume. It comprises five stages:

Stage 01: Exploratory data analysis. The raw MIMIC-IV tables are profiled for completeness, and the six core physiological features (lactate, white blood cell count, heart rate, respiratory rate, temperature, and age) are examined by sepsis status. Figure 4.2 shows the distribution of three features by sepsis status: septic stays exhibit higher lactate, faster heart rates, and elevated respiratory rates, but with substantial overlap that motivates multivariable modelling.

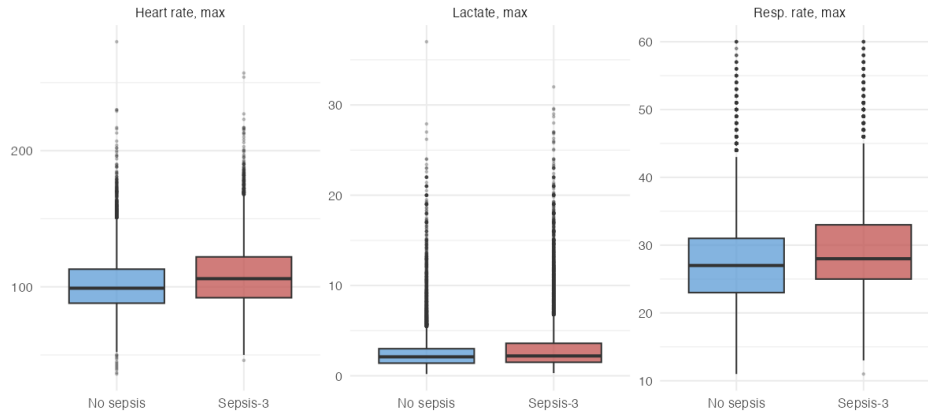


Figure 4.2: Boxplots of heart rate, lactate, and respiratory rate by sepsis status. All three features show genuine separation, but no single feature cleanly discriminates the groups.

Figure 4.3 shows the lactate distribution by sepsis status, confirming a right-skewed, overlapping pattern that informs the choice of log-transformation for distributional models.

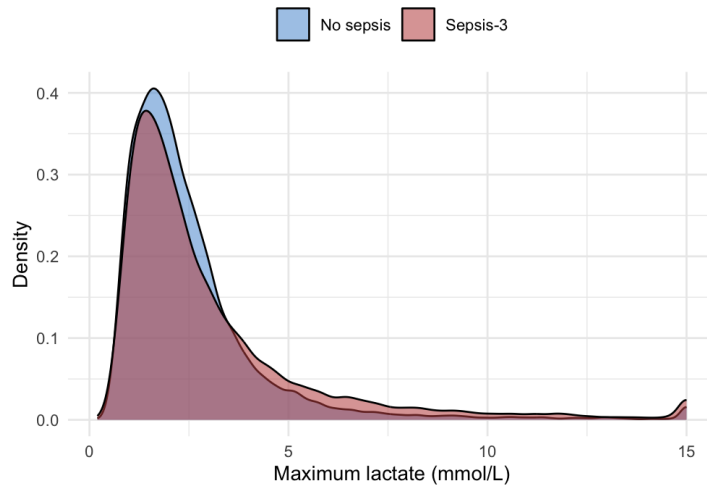


Figure 4.3: Kernel density estimate of maximum lactate by sepsis status. Both groups are right-skewed; the septic distribution is shifted upward but overlaps heavily with the non-septic distribution.

Stage 02: Cohort extraction. The adult, first-ICU-stay cohort of 65,241 stays is built from the administrative tables, applying the inclusion criteria of Section 2.6.

Stage 03: Suspected-infection detection. The prescription and microbiology tables are streamed to pair antibiotics with cultures inside the Sepsis-3 time windows, yielding a binary suspected-infection flag per stay.

Stage 04: Measurement extraction. The multi-gigabyte chart and laboratory tables are processed in bounded-memory chunks, extracting the core vitals and lab values and mapping them to clinical item identifiers.

Stage 05: SOFA computation and Sepsis-3 labelling. The six SOFA organ subsystems are scored from the first 24 hours of data, and the Sepsis-3 rule is applied to produce the binary sepsis label. The resulting analysis matrix has 65,241 rows, of which 37.1% meet Sepsis-3 criteria.

Stages 07–08: Hourly panel and onset labelling. Every stay is resampled onto a regular hourly grid (approximately 3.1 million stay-hours). Hourly SOFA is scored, and the exact onset hour is located. Figure 4.4 shows the distribution of onset hours, revealing a critical distinction: a large spike at hour 0 marks *prevalent* sepsis (already present at admission), while the right tail represents *incident* sepsis developing in the ICU, the target of early prediction.

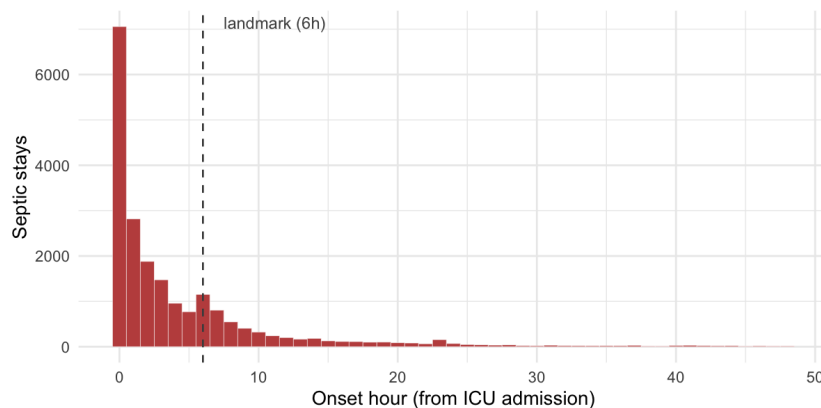


Figure 4.4: Distribution of sepsis-onset hour among septic stays. The spike at hour 0 marks prevalent sepsis; the right tail is incident sepsis developing in the ICU, which is the target for early prediction. The dashed line marks the hour-six landmark.

4.3 Model Layer (Scripts 09–19)

The model layer implements the seven statistical analyses. Because Analysis 2 (the dynamic landmark supermodel) is the central methodological contribution of this thesis, it receives the most detailed treatment here.

4.3.1 Analysis 1: Full-Feature Elastic-Net Nomogram

The Analysis-1 nomogram is an elastic-net penalised logistic regression at the hour-six landmark, using all 27 candidate features: vitals (6), severity scores (3), labs (4), six-hour trajectory slopes (4), binary “was measured” indicators (8, one per lab and per slope), plus age and sex. Nested five-fold cross-validation (outer folds for honest evaluation, inner folds for λ selection) yields out-of-sample predicted risks.

The key design choice is the **informative-missingness encoding**: each laboratory value is median-imputed and accompanied by a binary indicator recording whether it was measured. At hour 6, labs are missing in 70–88% of stays (clinicians order labs selectively in response to suspicion), so these indicators capture a genuine clinical signal rather than a nuisance pattern.

Figure 4.5 shows the clinical nomogram derived from the elastic-net-selected features, allowing point-based bedside risk estimation.

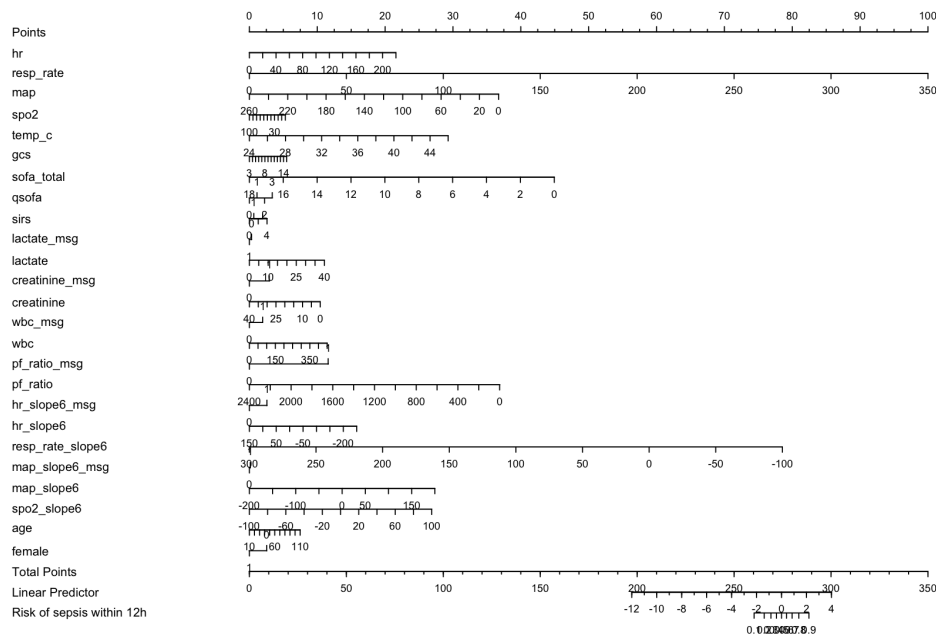


Figure 4.5: Nomogram for incident Sepsis-3 onset within 12 hours at the hour-six landmark. Each predictor contributes points on the top scale; the total is mapped to a predicted risk on the bottom scale. The “was-measured” indicators (`_msg` features) contribute substantial points, reflecting the informative-missingness signal.

4.3.2 Analysis 2: Dynamic Landmark Supermodel

A single landmark answers “predict from hour 6.” A deployed system must predict at *every* hour. The classical solution is a **landmark supermodel** [10]: the landmark construction is repeated on a grid of prediction hours $s \in \{6, 12, 18, 24, 36, 48\}$, the resulting datasets are stacked vertically, and a single penalised model is fitted with s , s^2 , and all 27 base-feature-by- s interactions (56 candidate features in total). Patient-grouped folds prevent information leakage across landmarks.

The supermodel design has three important properties:

1. **Parsimony.** One model replaces six separate per-landmark models, borrowing strength across prediction times while allowing the feature effects to

change smoothly with s through the interaction terms.

2. **Deployability.** At any ICU hour, the model is evaluated by setting s to the current hour and plugging in the patient’s current features. The output is a continuously updated risk estimate.
3. **Interpretability.** The coefficients are directly readable as log-odds ratios, and their dependence on s can be inspected through the interaction terms.

4.3.3 Analysis 3: Within-Patient Correlation

The hourly panel contains repeated measurements from the same patient, and these observations are strongly correlated. A naive pooled logistic model treats the approximately 76,000 at-risk stay-hours as independent, which produces artificially narrow confidence intervals. Analysis 3 addresses this in two ways:

1. A **GEE** with an exchangeable working correlation structure estimates population-averaged effects with cluster-robust sandwich standard errors.
2. A **ridge-penalised time-varying Cox model** on the counting-process formulation (start-stop intervals, one per hour) provides inferentially valid hazard ratios with cluster-robust variance estimation.

4.3.4 Analyses 4–7: Evaluation and Complementary Studies

Analysis 4 (Real-time evaluation). The supermodel is trained on 44,389 stays and evaluated on a held-out set of 6,000 stays (152,910 scored hours), computing the PhysioNet/CinC 2019 utility score and an alarm-burden sweep.

Analysis 5 (Subgroup fairness). Discrimination (AUROC), calibration-in-the-large, and error-rate parity (sensitivity and false-positive rate at a shared threshold) are stratified by sex and four age bands (<50, 50–64, 65–79, 80+) to assess equitable performance.

Analysis 6 (GBTM). A latent-class growth model [7] is fitted to first-24h SOFA trajectories via `flexmix`, selecting the number of classes by BIC. Class membership is linked to both mortality (Cox regression) and incident Sepsis-3 (logistic regression).

Analysis 7 (Causal analysis). The confounder-adjusted effect of early antibiotic administration (≤ 3 hours) on mortality is estimated using crude, PSM, IPW, and augmented-IPW estimators, with the adjustment set chosen by LASSO.

4.4 Evaluate Layer (Scripts 20–22)

The evaluate layer consolidates results across all analyses into a single comprehensive report and performs external validation. The frozen MIMIC model’s coefficients are serialised and applied to eICU-CRD by assembling the identical feature design matrix from eICU data and computing the linear predictor directly.

4.5 Implementation Notes

Extraction and panel assembly are implemented in Python with pandas; the statistical analyses and their reports are implemented in R. All intermediate artefacts are stored as Parquet (for large panel data) or CSV (for small result tables). All randomness uses a fixed seed (486649), and the entire pipeline is reproducible from a single driver script (`run_pipeline.py`) that executes the stages in order.

Chapter 5

System Evaluation

This chapter presents the experimental results of the pipeline, organised around the three sub-questions of Section 1.3. The dynamic landmark supermodel (Analysis 2) is the central contribution and receives the most detailed treatment. Internal MIMIC-IV results are presented first, followed by the external eICU-CRD validation and a critical discussion of strengths, limitations, and threats to validity.

5.1 Temporal Foundation: Onset Distribution and the Landmark Design

Before reporting model performance, two empirical facts about the data must be established, as they shape every subsequent result.

Prevalent versus incident sepsis. Of the 65,241 stays in the cohort, 37.1% satisfy the Sepsis-3 label on the first-24h analysis matrix (Section 2.5); an exact hourly onset time could then be located within the 72-hour panel for 32.0% of all stays (20,861 stays), which constitute the temporal cohort used from here on. Of these temporally labelled cases, 34% are already septic at ICU admission (onset hour 0). These prevalent cases are not predictable by any early-warning system. Conditioning at the hour-six landmark yields **48,829 at-risk stays**, of which 4,750 (9.7%) develop incident sepsis and 3,330 (6.8%) develop it within 12 hours. This is the correct, realistically imbalanced prediction target.

Time-to-onset survival analysis. Figure 5.1 shows the Kaplan-Meier cumulative incidence of sepsis onset from the hour-six landmark, stratified by SIRS status. Patients meeting ≥ 2 SIRS criteria at the landmark reach sepsis substantially faster (14.2% versus 7.8% by 24 hours after landmark; log-rank statistic 360, $p < 10^{-16}$).

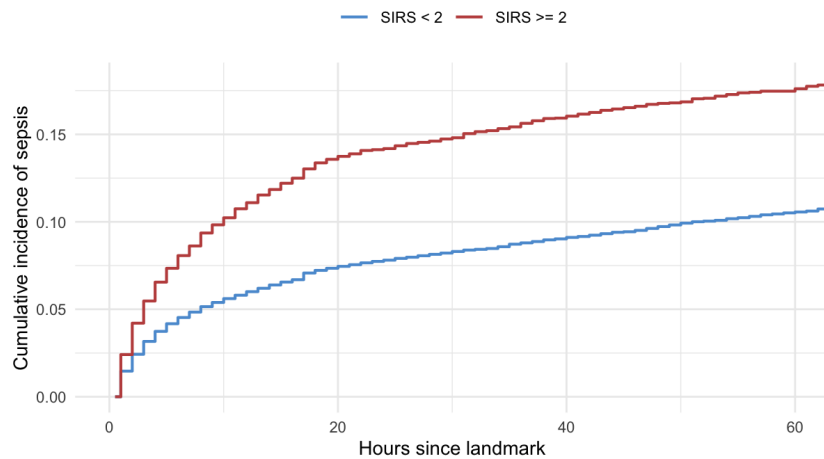


Figure 5.1: Cumulative incidence of sepsis onset from the hour-six landmark, stratified by SIRS status. Patients with ≥ 2 SIRS criteria develop sepsis faster, confirming that inflammatory markers carry predictive signal for the incident-onset task.

Competing risks. Death and discharge compete with sepsis onset: a patient who dies sepsis-free can never develop sepsis, so treating death as ordinary censoring overstates the onset incidence. Figure 5.2 shows the Fine-Gray cumulative incidence functions, confirming that SIRS raises the cumulative likelihood of onset even after accounting for the competing event (subdistribution hazard ratio for SIRS: 1.49).

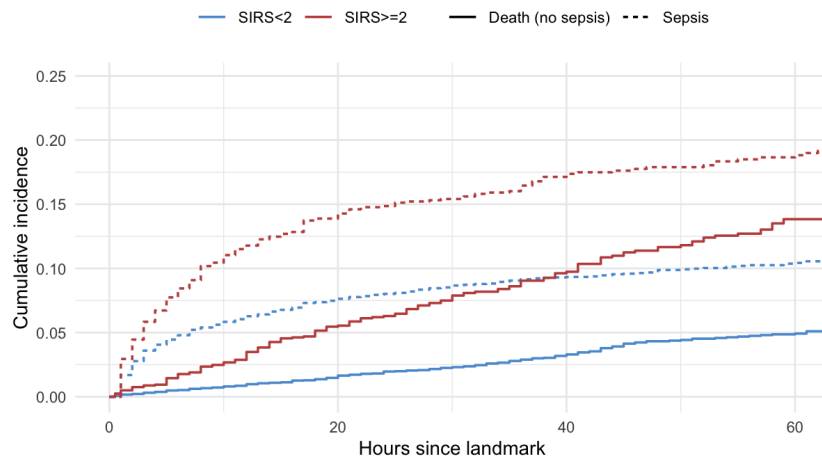


Figure 5.2: Cumulative incidence functions from the hour-six landmark accounting for death without sepsis as a competing risk, stratified by SIRS status. The solid lines show onset incidence; the dashed lines show the competing risk of death without sepsis.

5.2 Proportional-Hazards Testing and Time-Varying Cox (SQ1, C2)

A baseline Cox model estimates the hazard of onset from hour-six covariates. Table 5.1 reports the hazard ratios.

Term	HR	95% CI	p
Age	1.009	[1.007, 1.011]	$< 10^{-16}$
Heart rate	1.011	[1.009, 1.013]	$< 10^{-16}$
Resp. rate	1.021	[1.019, 1.023]	$< 10^{-16}$
MAP	0.988	[0.986, 0.990]	$< 10^{-16}$
SOFA total	0.762	[0.748, 0.777]	$< 10^{-16}$
SIRS	1.150	[1.104, 1.198]	2.5×10^{-11}

Table 5.1: Cox model for time to incident sepsis (hour-six landmark). SIRS and vital signs carry the hazard, while a single-timepoint SOFA value is *inversely* associated with incident onset.

A critical finding is that **SOFA at hour 6 is inversely associated** with incident onset (HR 0.76): patients with high organ dysfunction at hour 6 but no evolving infection tend *not* to cross the Sepsis-3 threshold later. This counterin-

tuitive result is clinically meaningful: it demonstrates that severity scores detect established dysfunction, not impending infection.

The global Schoenfeld residual test **rejects proportional hazards** ($p < 10^{-16}$), confirming on MIMIC-IV the non-proportionality that Kasal et al. [8] reported elsewhere (C2). In response, a **time-varying Cox model** is fitted on the counting-process formulation, updating covariates hour by hour. The time-varying model locates the hazard in *current heart rate* (HR 1.019 per bpm), while the single SOFA value remains inversely associated (HR 0.80): the predictive signal lives in *physiological dynamics*, not in a severity snapshot.

5.3 Temporal Structure: Trajectories and Auto-correlation

The cross-sectional analysis treated stays as independent observations. Within a patient, however, the hourly measurements form a genuine time series with strong serial dependence. Figure 5.3 shows the mean heart-rate trajectories by sepsis status (left panel) and the distribution of within-patient lag-1 autocorrelation of heart rate (right panel).

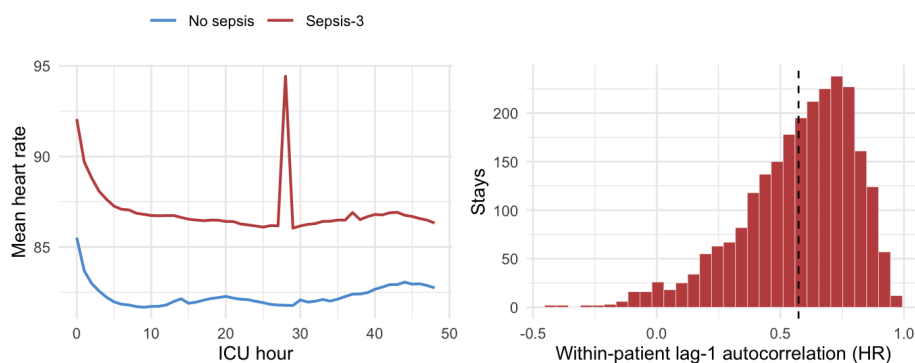


Figure 5.3: Left: mean hourly heart rate by sepsis status; septic patients run higher and diverge over time. Right: distribution of within-patient lag-1 autocorrelation of heart rate; the mean of 0.57 confirms strong hour-to-hour dependence that the static analysis discarded.

The mean within-patient lag-1 autocorrelation of heart rate is **0.57**, confirming that hourly observations within a patient are strongly dependent. This serial correlation is the statistical foundation that sequence models (including the landmark supermodel) are built to exploit: a model that conditions on the trajectory

of recent measurements captures information that a single-timepoint snapshot discards.

5.4 Benchmarking on the Incident-Onset Task (SQ3, C3)

Table 5.2 reports discrimination at the hour-six landmark for incident onset within 12 hours. The rule-based scores range from non-discriminative to inverse, whereas the statistical landmark models clearly exceed them.

Model / Score	AUROC
SOFA at hour 6	0.359 (inverse)
qSOFA at hour 6	0.492
SIRS at hour 6	0.586
Multivariable logistic (static, 6h)	0.749
Multivariable + 6h trajectory slopes	0.755
Full-feature elastic-net nomogram (Analysis 1)	0.775

Table 5.2: Discrimination at the hour-six landmark (incident onset within 12h) on MIMIC-IV. The full-feature elastic-net nomogram is the statistical ceiling at a single landmark.

This result answers SQ3 on internal data: **no single clinical score is useful** for incident sepsis. SOFA is even *inverse* (AUROC 0.36), because organ dysfunction without evolving infection does not progress to Sepsis-3. SIRS is the best single score (AUROC 0.59), consistent with its sensitivity to inflammatory signals. A multivariable combination of hourly measurements reaches AUROC 0.749, and adding six-hour trajectory slopes raises it to 0.755 (DeLong $p = 3.5 \times 10^{-7}$), supporting SQ2: the trajectory carries information a snapshot does not.

5.5 Analysis 1: Informative Missingness and the Nomogram (C4)

The full-feature elastic-net nomogram reaches AUROC **0.775**, a statistically significant gain over the 0.755 model (DeLong $p < 10^{-16}$). The most striking finding is *what* carries the signal: the ten largest coefficients are dominated by **missingness indicators**.

Feature	Coefficient	Odds Ratio
pf_ratio_msg (was P/F measured)	1.697	5.456
creatinine_msg	0.450	1.568
hr_slope6_msg	0.391	1.479
female	−0.388	0.678
sofa_total	−0.372	0.690
wbc_msg	0.306	1.358
temp_c	0.203	1.225
qsofa	0.160	1.174
sirs	0.105	1.110
gcs	0.074	1.076

Table 5.3: Ten largest coefficients (original scale) in the Analysis-1 elastic-net model. Features ending in _msg are binary “was measured” indicators. The act of ordering a P/F ratio is the single strongest predictor (OR 5.46), confirming an informative-missingness signal.

The P/F ratio measurement indicator alone has an odds ratio of **5.46**: having this test ordered multiplies the odds of subsequent onset by more than fivefold, independent of the test result. This finding quantifies a *label-recognition ceiling* [12]: the onset label depends on clinician suspicion (cultures and antibiotics), and the act of ordering labs reflects that same suspicion. This is a property of the data, not a model deficiency, and it sets an upper bound on what any retrospective model can achieve (C4).

Figure 5.4 shows the out-of-sample calibration curve for the Analysis-1 nomogram, demonstrating that predicted risks closely match observed onset rates.

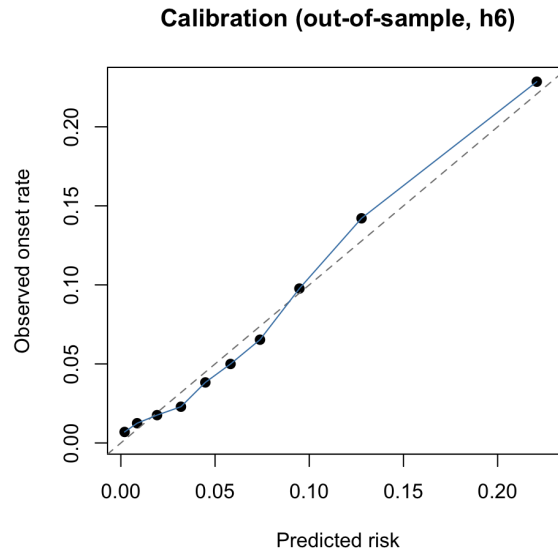


Figure 5.4: Calibration curve (out-of-sample) for the Analysis-1 elastic-net nomogram at the hour-six landmark. Points near the diagonal indicate that predicted risks are usable as probabilities, not merely as a ranking.

5.6 Analysis 2: Dynamic Landmark Supermodel (C1)

The Analysis-2 supermodel is the central deliverable: a single elastic-net logistic model fitted on 207,010 stacked landmark rows across 48,829 stays, with 56 features including landmark time s , s^2 , and 27 feature-by- s interactions. Patient-grouped five-fold cross-validation produces out-of-sample predictions at every landmark.

5.6.1 Overall Discrimination

The pooled out-of-sample AUROC across all landmarks is **0.797**. Table 5.4 shows the discrimination at each prediction hour, alongside a separate per-landmark model (no s terms, no pooling) as a comparator.

Landmark s (h)	At risk	Onset in 12h	AUROC super	AUROC separate
6	48,829	3,330	0.771	0.775
12	44,576	1,369	0.734	0.734
18	39,990	773	0.696	0.704
24	33,610	353	0.645	0.658
36	23,238	232	0.711	0.714
48	16,767	152	0.754	0.728

Table 5.4: Dynamic discrimination $AUC(s)$: one supermodel evaluated at each prediction hour, versus a separate model per landmark. The single supermodel matches the accuracy of bespoke per-landmark models while remaining one parsimonious, deployable object.

Figure 5.5 visualises these results.

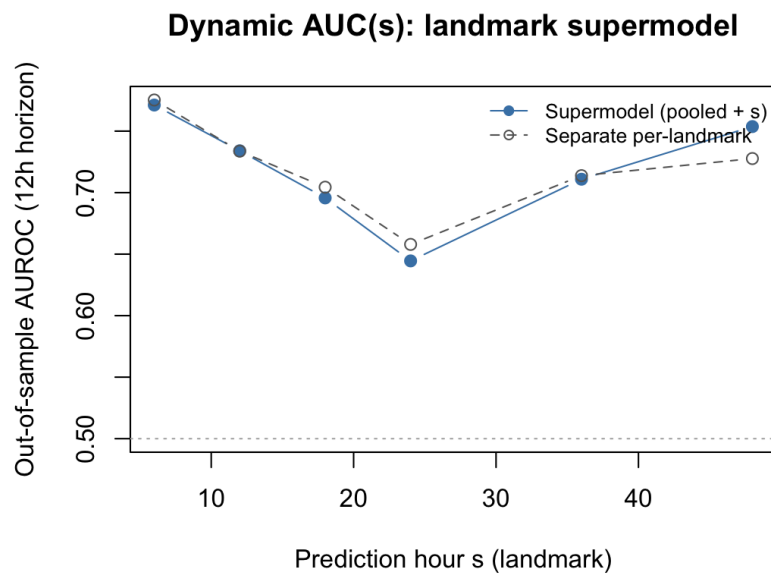


Figure 5.5: Dynamic $AUC(s)$: the supermodel's out-of-sample discrimination as a function of the hour at which the prediction is made, versus separate per-landmark models. The supermodel matches or exceeds the bespoke models at most landmarks while requiring only one set of coefficients.

5.6.2 Interpretation of the Dynamic Discrimination Pattern

The AUC pattern across landmarks is clinically informative. Discrimination is **highest early** (AUROC 0.77 at hour 6), when the pool of at-risk stays is largest and the onset signal is strongest. It **dips in the mid-stay window** (hours 18–24), where incident onset is rarest and hardest to predict among the patients who have survived the initial hours without developing sepsis. Discrimination then **recovers at later landmarks** (hours 36–48), because the patients still at risk at these hours constitute a selected subpopulation in which the remaining onset events carry a clearer signal.

This pattern is consistent with the epidemiology of incident sepsis: the highest-risk patients develop onset early, leaving a mid-stay population with a lower base rate and weaker signal, followed by a late-stay population where prolonged ICU exposure concentrates risk factors.

5.6.3 Time-Dependent AUC Across Horizons

Figure 5.6 shows the time-dependent AUC for predicting onset at increasing horizons from the hour-six landmark, using the cross-validated multivariable risk score. Discrimination holds above 0.70 across all clinically relevant horizons, confirming that the model provides useful risk stratification not only at the 12-hour window but also at longer prediction horizons.

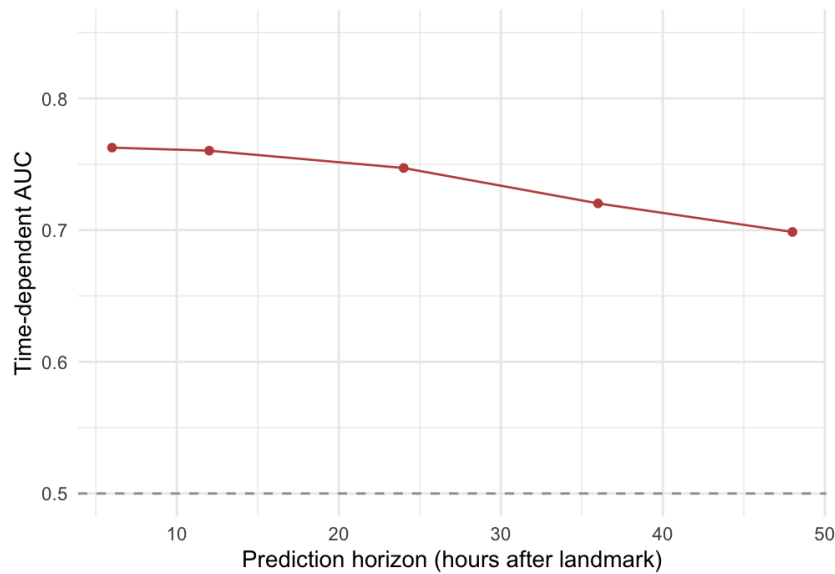


Figure 5.6: Time-dependent AUC for predicting sepsis onset over increasing horizons from the hour-six landmark. Discrimination remains above 0.70 across all tested horizons, demonstrating robust risk stratification over time.

5.7 Analysis 3: Within-Patient Correlation (GEE and Time-Varying Cox)

The within-patient exchangeable correlation estimated by GEE is $\alpha = 0.676$: hourly observations within a patient are strongly dependent, and the effective information is far below the 76,538 row count. Cluster-robust (sandwich) standard errors on the pooled logistic model inflate by up to **4.5×** compared to naive model-based standard errors (Table 5.5). Without this correction, a pooled GLM would report spuriously tight confidence intervals.

Term	OR	SE naive	SE robust	Inflation
(Intercept)	0.014	0.042	0.186	4.43×
z_hr	1.206	0.021	0.051	2.43×
z_map	0.880	0.021	0.047	2.24×
z_rr	1.137	0.019	0.043	2.26×
z_temp	1.083	0.020	0.046	2.30×
z_lac	0.845	0.030	0.099	3.30×
z_hour	0.175	0.043	0.193	4.49×

Table 5.5: Pooled logistic model: naive versus cluster-robust standard errors. Ignoring within-patient correlation understates every SE by 2–4.5 times.

The ridge time-varying Cox model on the counting-process formulation gives directionally consistent hazard ratios: heart rate (HR 1.30 per SD) and respiratory rate (HR 1.06) raise the hazard of onset, while MAP (HR 0.65) lowers it. These are inferentially valid companions to the predictive supermodel, not competitors to it.

5.8 Analysis 4: Real-Time Evaluation (Utility and Alarm Burden)

Discrimination alone is insufficient for a streaming alarm system. The supermodel is evaluated on a held-out 6,000-stay set (trained on the remainder, no leakage), scored at every hour from hour 6 to onset or discharge.

PhysioNet utility. The PhysioNet/CinC 2019 normalised utility score, which rewards early detection and penalises both late and false alarms, is maximised at **0.289** (0 = never-alarm baseline, 1 = per-hour oracle) at a risk threshold of 0.04. A positive score confirms that the model adds clinical value over silence on this incident-only ICU task.

Alarm burden. Figure 5.7 shows the sensitivity–alarm burden trade-off, the clinically decisive picture for deployment.

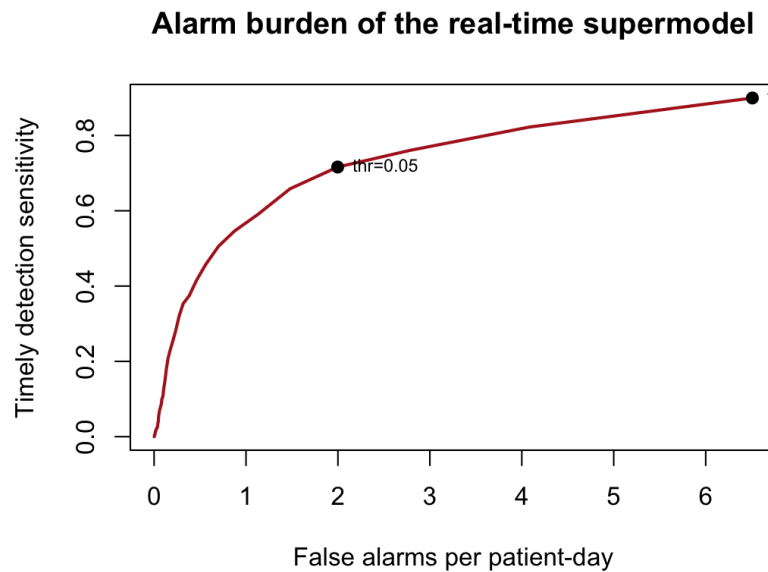


Figure 5.7: Alarm-burden trade-off for the real-time supermodel. High sensitivity ($\geq 90\%$) is achievable only at alarm rates (6.5 false alarms per patient-day) that would fatigue bedside staff. A practical operating point (51% sensitivity, 0.7 false alarms per patient-day) balances detection against alarm fatigue.

Table 5.6 summarises three operating points that illustrate the trade-off. Catching 90% of septic patients in a timely window costs about 6.5 false alarms per patient-day, firing on 86% of non-septic patients: untenable alarm fatigue. A calmer operating point (51% sensitivity) drops that to 0.7 per patient-day, alarming only 30% of non-septic patients. This is the number a deployment decision actually turns on.

Threshold	Sensitivity	FA/pt-day	Non-sep alarmed
0.02	0.90	6.5	86%
0.05	0.72	2.0	59%
0.09	0.51	0.7	30%

Table 5.6: Operating points: earlier/more-sensitive detection costs rapidly rising false-alarm burden.

5.9 Analysis 5: Subgroup Fairness

Fairness is assessed along three axes: discrimination parity (AUROC), calibration parity (calibration-in-the-large), and error-rate parity (sensitivity and false-positive rate at a single shared decision threshold). The threshold is fixed once on the full cohort at 50% overall sensitivity (predicted risk 0.116), then applied unchanged to every subgroup, so that any difference in error rates reflects the model rather than a different operating point. Table 5.7 reports all three axes.

Subgroup	n	Event rate	AUROC	Calib. ratio	Sens. [†]	FPR [†]
ALL	48,829	0.068	0.775	1.000	0.50	0.15
Female	21,519	0.063	0.762	1.003	0.42	0.12
Male	27,310	0.072	0.784	0.998	0.56	0.18
Age <50	9,680	0.058	0.756	1.056	0.40	0.11
Age 50–64	14,083	0.067	0.785	0.945	0.47	0.13
Age 65–79	15,960	0.069	0.777	0.985	0.51	0.16
Age 80+	9,106	0.079	0.769	1.050	0.61	0.22

Table 5.7: Subgroup fairness across three axes. Discrimination (AUROC 0.756–0.785) and calibration-in-the-large (ratio 0.945–1.056, perfect = 1.0) are equitable. [†]Sensitivity and false-positive rate are measured at a single shared threshold (0.116); at that fixed operating point the error rates track subgroup prevalence, exposing an equal-opportunity gap.

Two conclusions follow. On the first two axes the model is equitable: discrimination is stable (AUROC 0.756–0.785) and no subgroup is systematically over- or under-predicted (calibration ratio 0.945–1.056). On the third axis, however, applying one threshold to well-calibrated risks across groups with different base rates yields an *equal-opportunity gap*: sensitivity rises from 0.40 in the under-50 group to 0.61 in the over-80 group, and the false-positive rate from 0.11 to 0.22, with the same higher-prevalence groups (older patients, men) receiving both more true and more false alarms. This is a property of thresholding calibrated probabilities under unequal prevalence, not of miscalibration; a deployment would address it by an accepted single-threshold trade-off documented in advance or by group-aware thresholds, and the choice is a clinical governance decision rather than a purely technical one.

5.10 Analysis 6: Group-Based Trajectory Modelling

A latent-class growth model fitted to first-24h SOFA trajectories (2,438 stays with ≥ 12 hourly observations) selects **three classes** by BIC. Figure 5.8 shows the mean SOFA trajectory per class.

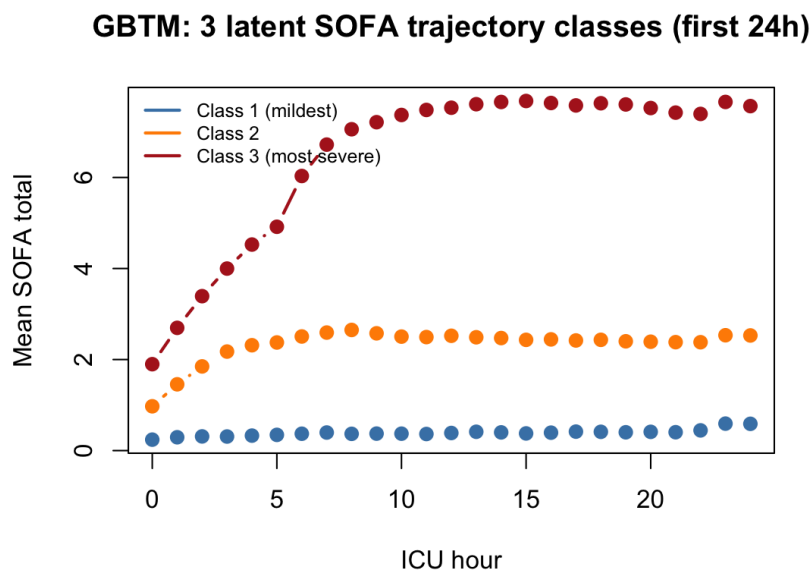


Figure 5.8: GBTM: three latent SOFA-trajectory classes over the first 24 ICU hours. Class 1 (mildest) is flat near zero; Class 2 (intermediate) rises to SOFA ~ 3 ; Class 3 (most severe) rises steeply to SOFA ~ 7 . The trajectory *shape*, not just its endpoint, carries prognostic and predictive information.

Class membership shows a strong, monotonic dose-response relationship with both outcomes. For in-hospital mortality, the most-severe class (Class 3) carries a hazard ratio of **5.019** ($p = 3.9 \times 10^{-8}$) relative to the mildest class, adjusted for age (Table 5.8). For incident Sepsis-3, the most-severe class carries an odds ratio of **18.81** ($p = 1.5 \times 10^{-74}$), confirming that the SOFA trajectory *shape*, not just its terminal value, is informative of subsequent onset (Table 5.9).

Term	Hazard ratio	<i>p</i> -value
Class 2 (vs. Class 1)	2.525	0.002
Class 3 (vs. Class 1)	5.019	3.9×10^{-8}
Age	1.029	1.7×10^{-9}

Table 5.8: Cox model: in-hospital mortality by GBTM trajectory class (reference = mildest class), adjusted for age.

Term	Odds ratio	<i>p</i> -value
(Intercept)	0.077	8.5×10^{-78}
Class 2 (vs. Class 1)	6.862	9.8×10^{-37}
Class 3 (vs. Class 1)	18.809	1.5×10^{-74}

Table 5.9: Logistic model: incident Sepsis-3 by GBTM trajectory class (reference = mildest class). The most-severe trajectory carries an 18-fold odds of subsequent onset.

This extends the Yang et al. [7] design, which linked trajectory classes only to mortality, by demonstrating that the trajectory shape also predicts the project’s own outcome of interest: incident sepsis.

5.11 Analysis 7: Confounder-Adjusted Causal Analysis

Every prior analysis reports an *association* (a risk score, a hazard ratio). Analysis 7 applies propensity-score matching (PSM), inverse-probability weighting (IPW), and doubly robust augmented-IPW estimation to estimate the confounder-adjusted effect of early antibiotic administration (≤ 3 hours of ICU admission) on in-hospital mortality among 24,203 Sepsis-3 patients (48% early, 18.7% mortality). The adjustment set is selected by LASSO from 11 candidate confounders; all 11 survive penalisation.

Figure 5.9 shows the covariate balance before and after propensity-score matching: every covariate’s standardised mean difference falls below the conventional 0.1 threshold after matching.

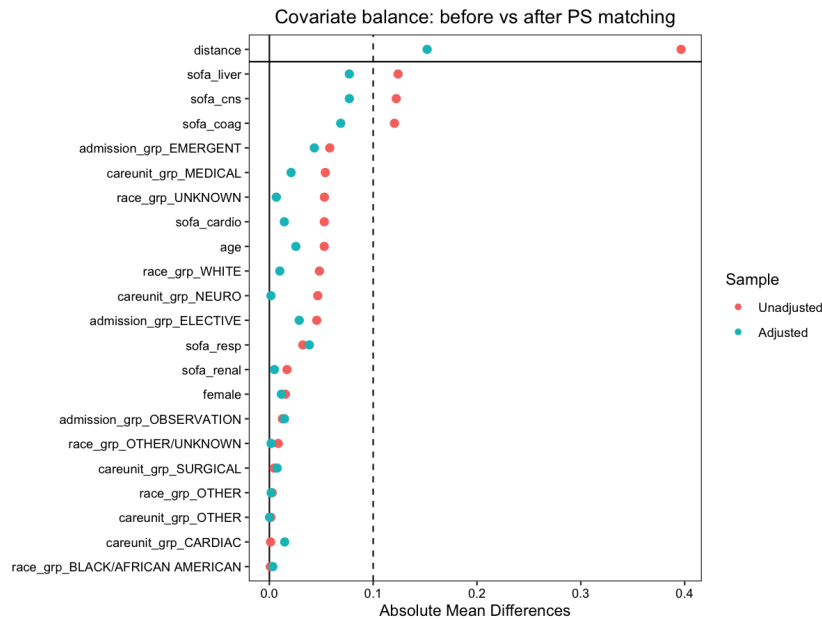


Figure 5.9: Love plot: covariate balance before (red) and after (teal) propensity-score matching. All standardised mean differences fall below 0.1 after matching, confirming adequate balance.

Table 5.10 summarises the effect estimates across all four estimators.

Method	Estimand	Odds ratio	95% CI
Crude (unadjusted)	OR	0.884	—
PSM (1:1 nearest)	OR (ATT)	0.899	[0.841, 0.962]
IPW (ATE weights)	OR (ATE)	0.889	[0.833, 0.950]
AIPW / doubly robust	RD	−0.019	[−0.028, −0.011]

Table 5.10: Effect of early (≤ 3 h) versus delayed antibiotics on in-hospital mortality. All four estimators agree in direction and magnitude: early antibiotics associate with approximately 10% lower odds of death.

Agreement across four estimators built on different assumptions provides the strongest evidence this observational pipeline can offer for a real association: early antibiotics are associated with lower mortality. However, as an observational analysis, it cannot rule out unmeasured confounders.

5.12 External Validation on eICU-CRD (SQ3, C5)

The frozen Analysis-1 nomogram was applied without any refitting to **113,597 at-risk eICU stays** across more than 200 US ICUs (4,782 incident onsets within 12 hours of the hour-six landmark). Table 5.11 reports the frozen-transport performance alongside the MIMIC development performance and an internal eICU refit ceiling.

Metric	eICU	MIMIC
AUROC (frozen transport)	0.686	0.775
95% CI	[0.678, 0.694]	—
AUROC (internal eICU refit ceiling)	0.711	—
Calibration slope	0.54	1.00
Mean predicted vs. observed risk	0.121 / 0.042	—
Brier (frozen → recalibrated)	0.052 / 0.040	0.059

Table 5.11: External validation of the frozen Analysis-1 nomogram on eICU-CRD. Discrimination transports with modest degradation; the gap between frozen (0.686) and internal refit (0.711) reflects population heterogeneity, not model failure.

Figure 5.10 shows the frozen and recalibrated calibration curves on eICU. The frozen model over-predicts risk approximately threefold (calibration slope 0.54, reflecting the lower onset base rate in the multi-centre population), but a single-parameter intercept-and-slope recalibration corrects this, reducing the Brier score from 0.052 to 0.040.

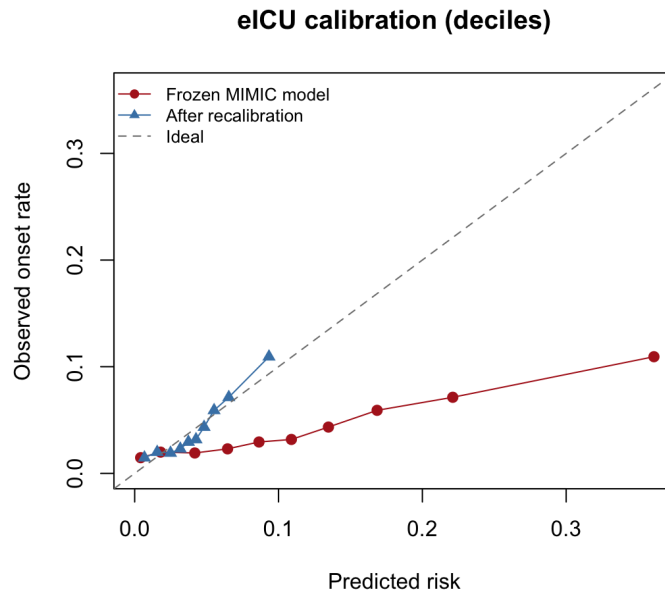


Figure 5.10: eICU calibration by risk decile: the frozen MIMIC model (red) over-predicts, and a single-parameter recalibration (blue) restores agreement with observed onset rates. The dashed grey line marks perfect calibration.

Interpretation. The frozen-transport AUROC (0.686) sits close to the eICU-native refit ceiling (0.711), so most of the drop from MIMIC’s 0.775 reflects a harder, more heterogeneous multi-centre population rather than model failure. The DeLong test between frozen and refit is significant ($p < 10^{-27}$), confirming a modest but real gap that could be closed by local recalibration. This is the expected result for any single-centre model transported to 200+ ICUs, and it is consistent with the validation degradation reported by Wang et al. [9] across the broader literature.

5.13 Cross-Sectional Foundational Analysis

Before the temporal landmark models, a cross-sectional statistical study established whether routine physiology that is *not* part of the sepsis definition can identify prevalent Sepsis-3, and provided the methodological foundation on which the later analyses build. Six deliberately non-circular predictors are used—lactate, white-cell count, heart rate, respiratory rate, temperature, and age—each lying outside the SOFA/Sepsis-3 definition, so the model recovers the label from independent signal rather than from the quantities used to construct it. After a

physiological plausibility filter and complete-case restriction, 36,252 stays remain (50% Sepsis-3, an enriched sample because lactate is ordered preferentially in sicker patients).

Individually, each feature carries only a small effect: the standardised difference in maximum lactate between groups is a Cohen’s d of 0.21, the white-cell rank-biserial correlation is 0.11, and the association between tachypnoea ($\geq 22/\text{min}$) and sepsis has a Cramér’s V of 0.07. Two unsupervised two-component Gaussian mixtures fitted to log-lactate and to log-white-cell count recover sepsis status at only 54% and 49% respectively, close to chance, confirming that no single marker separates the groups. Combining the six predictors in a logistic model, however, yields fair discrimination. Table 5.12 reports the cross-sectional prediction scores.

Model / metric	Value
Lactate-only logistic (10-fold CV AU-ROC)	0.540
Six-predictor logistic (10-fold CV AU-ROC)	0.665 [0.660, 0.671]
Six-predictor calibration slope	1.00
Six-predictor Brier (vs. null)	0.229 / 0.250
Six-predictor PR-AUC (prevalence base-line)	0.676 (0.500)

Table 5.12: Cross-sectional discrimination of prevalent Sepsis-3 from non-circular physiology. The six-predictor model is well calibrated (slope 1.00) and clearly exceeds lactate alone, but its ceiling is modest because the predictors exclude the severity scores and the target is cross-sectional prevalent classification, not incident onset.

This cross-sectional ceiling (AUROC 0.665) is not directly comparable with the landmark nomogram of Analysis 1 (AUROC 0.775): the landmark model adds the severity scores and the informative-missingness indicators, and it targets the harder incident-onset task on a realistically imbalanced cohort. The comparison is instructive precisely because it shows what the temporal design adds. A first sign of the informative-missingness effect already appears here: Sepsis-3 prevalence is 49.8% among stays with a measured lactate versus 18.7% among those without ($\chi^2 p < 10^{-16}$), foreshadowing the label-recognition ceiling quantified in Analysis 1 (C4).

Two diagnostic figures from this analysis support modelling choices carried into the pipeline. Figure 5.11 shows the GAM-estimated spline for lactate’s effect on prevalent Sepsis-3, confirming a monotone but non-linear relationship that justifies the log-transformation used throughout. Figure 5.12 shows the precision-

recall curve and decision-curve analysis for the six-predictor cross-sectional model, demonstrating net clinical benefit above both “treat all” and “treat none” strategies across a clinically relevant range of threshold probabilities.

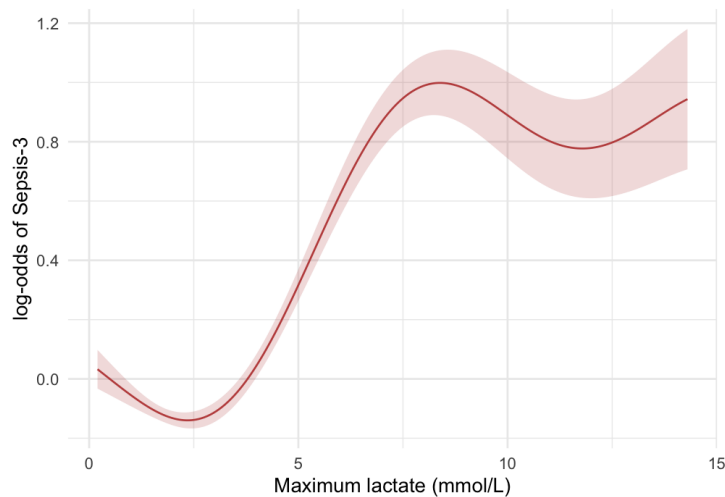


Figure 5.11: GAM spline: the non-linear relationship between lactate and the log-odds of prevalent Sepsis-3. The effect is monotone-increasing but saturates at high lactate values, supporting the log-transformation used in the linear models.

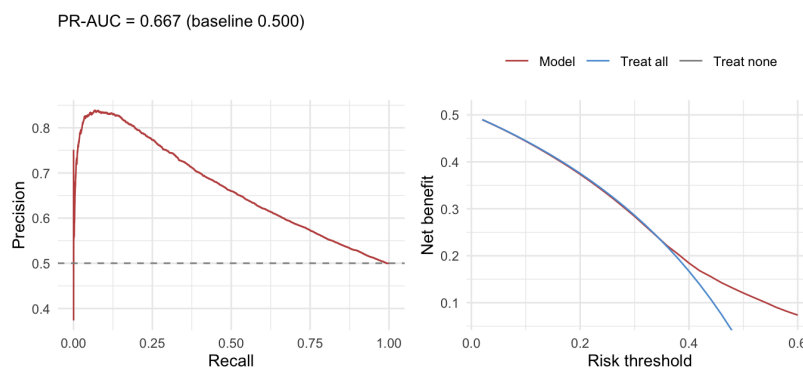


Figure 5.12: Left: precision-recall curve for the six-predictor cross-sectional model (PR-AUC shown). Right: decision-curve analysis demonstrating net benefit above both “treat all” and “treat none” across threshold probabilities from 0.02 to 0.20.

5.14 Discussion and Threats to Validity

5.14.1 Central Finding

The central empirical finding of this thesis is that the ceiling on incident-onset prediction is set by the *label's dependence on clinician recognition* (when cultures and antibiotics are ordered) rather than by model class. This is why the strongest predictors are missingness indicators and why a purely statistical model is competitive with more complex approaches: no neural network is required to approach a ceiling imposed by the labelling process itself. This result aligns with, and sharpens, the informative-missingness literature [12], [13].

5.14.2 Strengths

1. **Interpretability.** Every coefficient is a readable log-odds ratio or hazard ratio. Clinicians can trace the model's reasoning, and regulatory frameworks (e.g., the EU AI Act) favour transparency.
2. **Assumption testing.** The proportional-hazards assumption was tested and, because it failed, a time-varying reformulation was fitted in response. This workflow is absent from the reviewed MIMIC-IV literature.
3. **External validation.** The frozen-transport paradigm (no refitting) is the most honest form of external validation and directly tests the model's deployability.
4. **Clinical framing.** The alarm-burden analysis translates discrimination into the metric that deployment decisions actually depend on.

5.14.3 Threats to Validity

Several threats remain and are documented rather than hidden:

- The onset label depends on documentation timing (when cultures and antibiotics are ordered); “onset” partly reflects clinical recognition rather than biological transition.
- The assumed-zero SOFA baseline may slightly overcount organ dysfunction in patients with chronic organ impairment.
- qSOFA uses mean arterial pressure (MAP) as a proxy for systolic blood pressure, which is not the original qSOFA definition.

- The time-varying Cox and GBTM analyses use sampled subsets (2,254 and 2,438 stays respectively) for computational tractability.
- The external validation on eICU-CRD requires an adaptation in the suspected-infection trigger (antibiotics alone, with culture refinement when available) due to eICU's limited microbiology capture ($\sim 1.5\%$ of stays).

Chapter 6

Conclusion

This thesis set out to answer whether a purely statistical, interpretable model could predict impending Sepsis-3 onset ahead of clinical recognition on MIMIC-IV, matching or exceeding established rule-based scores and surviving external validation. The motivation was a specific imbalance in the literature: statistical work on MIMIC-IV overwhelmingly predicts mortality after diagnosis, rarely tests the proportional-hazards assumption, and seldom validates onset-prediction models externally.

The work delivered an interpretable pipeline whose central deliverable is a dynamic landmark supermodel (C1): a single elastic-net logistic model fitted on 207,010 stacked landmark rows across 48,829 stays, producing a pooled out-of-sample AUROC of **0.797** across six prediction hours. The supermodel matches or exceeds bespoke per-landmark models while remaining one parsimonious, deployable object that emits continuously updated hourly risk. The pipeline formally tests and acts on the proportional-hazards assumption (C2), reframes the benchmark by showing that bedside severity scores fail on the incident-onset task while a multivariable nomogram reaches an AUROC of 0.775 (C3), characterises an informative-missingness ceiling in which whether a laboratory test was ordered outweighs its value (C4), and, closing the validation gap, transports the frozen model to eICU-CRD (C5). External discrimination held at 0.686 against an internal eICU ceiling of 0.711, and the expected miscalibration was corrected by a single-parameter recalibration.

The central lesson is that, for retrospectively labelled ICU data, the ceiling on onset prediction is set by the labelling process rather than by model complexity; an interpretable statistical model is therefore a scientifically honest and clinically transparent choice. Avenues for further research follow naturally. The same landmark substrate can be used to benchmark machine-learning approaches against this interpretable baseline on identical splits; the supermodel can be transported across all landmark hours rather than the hour-six landmark alone; and prospec-

tive, multi-centre evaluation would test whether the recalibrated model supports clinically useful early warning at an acceptable alarm burden. By pairing rigorous internal methodology with a genuine external test, this work offers a reproducible and transportable baseline for the next generation of sepsis early-warning research.

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